

Martin Morales

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OVERVIEW

I am a software developer driven by a passion for solving real-world, complex problems through abstraction and code. Since high school, I have approached challenges with the belief that most problems can be decomposed and addressed through thoughtful software design. This mindset motivates me to continuously build and publish projects on GitHub that solve problems relevant to my own experiences. I enjoy selecting appropriate technologies and approaching problems from multiple perspectives, with experience across frontend and backend development, databases, AI, system design, and data analysis.

SKILLS

Programming: Python, JavaScript/TypeScript, Go, C/C++, C#, Java

Data & Analytics: Data Analysis, Machine Learning, SQL, Data Modelling, Dashboarding

Web Development: React, Next.js, Vue.js, Angular, HTML, CSS, Tailwind, Bootstrap

Backend & Databases: .NET, Node.js, Django, SQL Server, MongoDB, Firebase

Cloud & DevOps: AWS, GCP, Docker, CI/CD, Git/GitHub

Software Engineering: AI Development, Agentic Development, System Design, APIs, Data Structures & Algorithms

Other: Problem-Solving, Team Leadership, Technical Communication, Research & Publication

EXPERIENCE

Associate Software Developer - SOTI Inc.

June 15, 2026 - Present

- Designs and builds an internal evaluation framework in Angular and C# to benchmark the performance of open-source and proprietary LLMs across internal use cases, informing model selection and deployment decisions.

Research Assistant - Faculty of Business Administration. Lakehead University

September 15, 2024 - Present

- Built an end-to-end data pipeline to scrape and analyze social media discourse on menopause and ADHD in the workplace, applying sentiment analysis and topic modeling to surface key discussion themes and workplace attitudes.
- Co-authored a [peer-reviewed research article](#) published in Frontiers in Psychology, leading data collection, analysis, and visualization, and authoring the Methods and Ethics sections.

Data Analyst - Lakehead University International Enrolment

May 1, 2023 - December 22, 2025

- Developed and tested multiple machine learning models with Scikit-learn and TensorFlow that predict student enrollment given factors like engagement with marketing campaigns, country, level of study, etc.
- Designed and implemented a historic enrollment analysis dashboard by integrating 7 years of applicant and enrollment data, enabling stakeholders to track key performance indicators (KPIs), identify trends, and evaluate cyclical performance for data-driven decision making.
- Engineered Python and Google Apps Script automation tools to clean and import operational data (prospective, applicant, agents, etc.), significantly improving data accuracy and productivity.
- Supervised and mentored multiple team members in advanced data cleaning and importing procedures.

PROJECTS

Lorcana Engine — Open Source Contributor

March 2026 – Present

<https://github.com/TheCardGoat/lorcana-engine>

- Contributed to an open-source browser-based simulator for the Disney Lorcana TCG, improving UI interactivity and fixing bugs with card interactions.

Reddit r/Canada AI Explorer (AI Collective Hackathon 2nd Place)

January 2026

<https://github.com/Ajambot/ai-collective-hackathon>

- Integrated Google Gemini API to generate contextual AI analysis of Reddit data, designing prompts to produce structured, insight-driven summaries from retrieved content.
- Designed and implemented an end-to-end data pipeline to ingest, normalize, and serve large volumes of Reddit data using React and Node.js, with metadata linking to map AI-generated insights back to original records for auditability.
- Owned the full project lifecycle — system design, implementation, testing, and optimization — under hackathon time constraints, including data access improvements to reduce redundant computation.

HTTP/1.1 Server & Load Balancer Implementation in Go

August - September 2025

<https://github.com/Ajambot/Go-HTTP-Server-1.1>

- Implemented a fully functional HTTP/1.1 server from scratch in Go, adhering to RFC 9110 and RFC 9112.
- Built a load balancer in Go implementing multiple scheduling algorithms (e.g., round-robin, least-connections, weighted) with concurrent request handling to distribute traffic across backend servers.

Riposte

January - April 2025

<https://github.com/Ajambot/scheduler>

- Led a team of developers in building an AI-assisted social media scheduling app using Next.js, integrating with Bluesky and X (formerly Twitter).
- Developed and implemented the Bluesky API connector to support login/logout, post creation, fetching user posts, and session management.
- Integrated Google Gemini LLM to generate personalized post captions based on user history and intent.

Quoteversation

February - March 2024

<https://github.com/Ajambot/Quoteversation>

- Employed the MERN stack and Typescript to develop an X/Twitter/Threads clone that allows users to share their favourite quotes with their friends and followers.
- Deployed the app on AWS using an EC2 instance for the backend with an Nginx Reverse Proxy handling https requests, and an S3 Bucket in collaboration with Cloudfront to serve the app to end users.
- Containerized the app using Docker for ease and reliability of deployment and set up Github Workflows for Continuous deployment to the AWS services.

WhisperLink

January - April 2024

<https://github.com/Ajambot/WhisperLink>

- Collaborated in a team of 3 developers to design, plan and implement an online chat app focused on anonymity and security.
- Spearheaded the implementation of end-to-end encryption, and seamless backend connectivity using React and Firebase.
- Co-authored and presented a peer-reviewed research paper at IEEE SNPD 2024, demonstrating rigorous analytical methodology.
 - o DOI: <https://doi.org/10.1109/SNPD61259.2024.10673917>

EDUCATION

Lakehead University

September 2021 - December 2025

HBSc, Computer Science with Business Focus. Minor in Economics. Co-op.

Thunder Bay, ON

- **Cumulative GPA:** 91.7%
- **1st Place** Lakehead IEEE Hackathon, **2nd Place** AI Collective Hackathon
- **President's list:** 2021-2022, 2022-2023, 2023-24, 2024-2025
- **Awards:** Ontario Remembrance Award 2024 (10,000 CAD), Sher Ali Mirza Award 2022 (1,780 CAD)
- **Undergraduate Research:** See WhisperLink Project and Research Assistant above